Sessional Primitives

Course Number: CSE 206
Course Title: Digital Logic Design Lab
Credit Hour: 1.5
Contact Hour per Week: 3
Server Needed (If any): No
Additional Requirements (if any): No

Themes to Be Covered:

Experiment 1:
Title: Implementing circuit with basic gates

Outline (within 50-100 words): This lab introduces students to basic logic gates and ICs. Students learn how to implement Boolean functions in a digital circuit using logic gates.

Online Task: Implementation of (some) three Boolean functions using logic gates.

Offline Task: Preparation of report (1 report/group) on the experiment performed. The report contains the followings:

- _Problem Specification.
- _Required Instruments.
- _Truth Table.
- _Circuit diagram with pin number.
- _Overall discussion on the experiment.

AND
OR
NOT
Report

Mode of Evaluation (Presentation, Quiz or Viva):

- Testing and verification of the circuit implemented by each group for the desired input/output combinations.
- Report writing.

Suggested Percentage Weight:

- For successful completion of the experiment (each group member will get): 30/Total number of experiments performed.
- For submitted report: 10/Total number of experimentsperformed.

Experiment 2:
Title: Truth tables and simplification using Boolean algebra.

Outline (within 50-100 words): In this lab students simplifies a given Boolean function and implements the simplified function in a digital circuit that results in optimum number of gate usage.

Online Task: Simplification of given Boolean functions and implementation using logic gates.

Offline Task: Preparation of report (1 report/group) on the experiment performed. The report contains the followings:

- Problem Specification.
- Required Instruments.
- Truth Table.
- Circuit diagram with pin number.
- Answer to some given questions related to the experiment.
- Overall discussion on the experiment.

Mode of Evaluation (Presentation, Quiz or Viva):

- Testing and verification of the circuit implemented by each group for the desired input/output combinations.
- Report writing

Suggested Percentage Weight:

- For successful completion of the experiment (each group member will get): 30/Total number of experiments performed.
- For submitted report: 10/Total number of experiments performed.

Experiment 3:

Title: Truth tables and K-maps.

Outline (within 50-100 words): In this lab students transform a problem description to an equivalent Boolean function, then simplify it using K-map and finally implement the function in a digital circuit.

Online Task: Find the Boolean function that corresponds to a given problem specification, simplify it using K-map and then implement using logic gates.

Offline Task: Preparation of report (1 report/group) on the experiment performed. The report contains the followings:

- Problem Specification.
- Required Instruments.
- Truth Table.
- Circuit diagram with pin number.
- Answer to some given questions related to the experiment.
- Overall discussion on the experiment.
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Mode of Evaluation (Presentation, Quiz or Viva):

- Testing and verification of the circuit implemented by each group for the desired input/output combinations.
- Report writing

Suggested Percentage Weight:

- For successful completion of the experiment (each group member will get): 30/Total number of experiments performed.
- For submitted report: 10/Total number of experiments performed.

Half Adder
Full Adder
H-S
F-S

Experiment 6:
Title: Arithmetic circuit design.

Outline (within 50-100 words): In this lab students learn how to design a digital circuit that performs useful arithmetic operations such as addition and subtraction.

Online Task: Design and implementation of digital circuits that perform comparison and subtraction operation.

Offline Task: Preparation of report (1 report/group) on the experiment performed. The report contains the followings:

- Problem Specification.
- Required Instruments.
- Truth Table.
- Circuit diagram with pin number.
- Answer to some given questions related to the experiment.
- Overall discussion on the experiment.
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Mode of Evaluation (Presentation, Quiz or Viva):

- Testing and verification of the circuit implemented by each group for the desired input/output combinations.
- Report writing

Suggested Percentage Weight:

- For successful completion of the experiment (each group member will get): 30/Total number of experiments performed.
- For submitted report: 10/Total number of experiments performed.

Experiment 7: (a), 7(b)
Title: Circuit design using IC 7483.

4-bit full Adder

Outline (within 50-100 words): In this lab students learn how to use a commercially available adder IC 7483 to perform arithmetic operations.

Online Task: Implementation of given arithmetic operations (addition and subtraction) using 7483.

Offline Task: Preparation of report (1 report/group) on the experiment performed. The report contains the followings:

- Problem Specification.
- Required Instruments.
- Truth Table.
- Circuit diagram with pin number.
- Answer to some given questions related to the experiment.
- Overall discussion on the experiment.

Look ahead carry

Mode of Evaluation (Presentation, Quiz or Viva):

- Testing and verification of the circuit implemented by each group for the desired input/output combinations.
- Report writing

Suggested Percentage Weight:

- For successful completion of the experiment (each group member will get): 30/Total number of experiments performed.
- For submitted report: 10/Total number of experiments performed.

(c) BCD
(d) Comparator
(e) parity checker